C32 NMR Spectroscopy – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: NMR Basics **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Standard used in NMR				TMS, $Si(CH_3)_4$
2	Why TMS is used				Single peak; inert; volatile
3	Chemical shift of TMS				0 ppm
4	Symbol and units of chemical shift				δ, ppm
5	Solvent used for NMR				$CDCl_3$
6	Why a deuterated solvent is used				Gives no ^{1}H signal
7	Number of peaks in ^{13}C NMR shows				Number of carbon environments
8	Number of peaks in ^{1}H NMR shows				Number of hydrogen environments
9	Integration ratio shows				Relative number of H in each environment
10	Radiation used in NMR				Radio waves

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Splitting **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	*n* + 1 rule				Peak splits into *n* + 1, where *n* = H on adjacent carbons
2	Singlet				0 H on adjacent carbons
3	Doublet				1 H on adjacent carbons
4	Triplet				2 H on adjacent carbons
5	Quartet				3 H on adjacent carbons
6	Pattern for an ethyl group				Triplet (3H) + quartet (2H)
7	Splitting in ^{13}C NMR spectra				None
8	Pattern for an isolated CH_3 (e.g. next to C=O)				Singlet
9	Where chemical shift values are found				Data booklet
10	Use of NMR in medicine				MRI scanning

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.